

The Dispersion Calculator

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1 What the calculator computes

The Dispersion calculator describes the short-range van der Waals geometry around a nucleus, restricted in this implementation to nearby aromatic ring vertices. London dispersion is the part of the intermolecular interaction that comes from correlated electronic fluctuations. In a pair-potential formula it is often represented by an attractive energy proportional to $-C_6/r^6$, where r is the separation and C_6 carries the species-dependent polarizability information.

The code does not compute that energy. It keeps the r^{-6} distance law and the direction from a ring atom to the target nucleus, sets the dispersion coefficient to one, and records the resulting geometry as a rank-2 tensor with units $\text{\AA}^{-6}$. “Rank-2” here means that the quantity has two Cartesian indices: entry ab says how the geometry distinguishes an output direction a from an input direction b . The calculator also stores a separate positive scalar r^{-6} contact strength, again with units $\text{\AA}^{-6}$.

This bears on NMR shielding because nearby polarizable electrons change the electronic response of the observed nucleus to an applied magnetic field. The calculator does not solve that electronic response. Its output is a geometric descriptor that can correlate with a DFT shielding contribution after calibration. Before calibration it is not a chemical shift and not a shielding in ppm.

2 Where shielding enters

For a nucleus in an applied magnetic field B_0 , the local field is written

$$B_{\text{nuc}} = (I - \sigma)B_0. \quad (1)$$

Here I is the 3×3 identity matrix and σ is the nuclear shielding tensor. In physical NMR, σ is dimensionless: it is the linear-response coefficient connecting the applied field to the field screened by the electrons. The matrix is needed because a molecule is not spherically symmetric. A field applied along x can induce a secondary field with x , y , and z components at the nucleus, and rotating the molecule changes those components.

Equation (1) is a quantum-mechanical statement about electrons. The external field perturbs the electronic state; the perturbed electrons generate their own magnetic field at the nucleus; the part linear in B_0 is recorded in σ . The isotropic chemical shift reported in solution NMR is obtained from the orientationally averaged shielding after reference subtraction and sign convention. That scalar observable is connected to the tensor, but it does not contain all the tensor information.

The dispersion model used here isolates one geometric source of shielding variation: how close aromatic ring atoms sit to the observed nucleus, and from which directions they approach it. Aromatic rings are treated as sets of vertices at their ring-atom coordinates. Each accepted vertex contributes a short-range tensor whose magnitude falls as r^{-6} . The result is not a London dispersion energy and not an electronic-structure calculation. It is the orientation-resolved geometry of nearby van der Waals contacts, left for calibration to connect to the DFT shielding tensor.

3 The tensor split used by the code

The C++ type `SphericalTensor` decomposes any real 3×3 tensor X into three pieces. The scalar piece is

$$T_0 = \frac{1}{3} \text{tr } X, \quad (2)$$

where $\text{tr } \mathbf{X} = X_{xx} + X_{yy} + X_{zz}$. This is the isotropic part, the part that remains when all molecular orientations are averaged.

The antisymmetric part is stored as the three-component vector

$$\mathbf{T}_1 = \frac{1}{2} \begin{pmatrix} X_{yz} - X_{zy} \\ X_{zx} - X_{xz} \\ X_{xy} - X_{yx} \end{pmatrix}. \quad (3)$$

This vector is a compact storage form for the part of the matrix that changes sign under transpose. It is usually not measured directly in standard isotropic solution NMR, but the full nine-number layout includes it.

The remaining part is the symmetric traceless matrix

$$\mathbf{S} = \frac{\mathbf{X} + \mathbf{X}^T}{2} - T_0 \mathbf{I}, \quad \text{tr } \mathbf{S} = 0. \quad (4)$$

It has five independent entries. The code packs them as

$$\mathbf{T}_2 = \begin{pmatrix} \sqrt{2} S_{xy} \\ \sqrt{2} S_{yz} \\ \sqrt{3/2} S_{zz} \\ \sqrt{2} S_{xz} \\ (S_{xx} - S_{yy})/\sqrt{2} \end{pmatrix}. \quad (5)$$

These factors are the normalization in `Types.cpp`. They preserve the Frobenius length: $\sum_m T_{2m}^2 = \sum_{ab} S_{ab}^2$. The packed order is

$$[T_0, T_{1x}, T_{1y}, T_{1z}, T_{2,-2}, T_{2,-1}, T_{2,0}, T_{2,+1}, T_{2,+2}].$$

For this calculator, the tensor contribution from each accepted vertex is symmetric and traceless by construction. Its total therefore has $T_0 = 0$ and $\mathbf{T}_1 = \mathbf{0}$, up to floating-point roundoff, and its tensor signal lives in the five T_2 components. The separate scalar contact sum is not the trace of that tensor. It is the stored isotropic r^{-6} contact strength used as a separate feature.

4 The method implemented

Let target atom i have position $\mathbf{r}_i$, in Å. Let an aromatic ring R have vertex positions $\mathbf{v}_j$, also in Å, one for each ring atom. The code first queries ring centers within 15.0 Å, using the shared ring-current spatial cutoff from `data/calculator_params.toml`. This broad query only selects candidate rings. The actual dispersion range is enforced per vertex by the 5.0 Å cutoff below.

For each candidate ring, two ring-level exclusions are applied. The near-field test rejects a target whose distance from the ring center is not greater than the ring radius. In the code this filter uses the ring diameter as the source extent and the parameter `near_field_exclusion_ratio` = 0.5. The topology test rejects the whole ring if the target atom is a ring vertex or is covalently bonded to any ring vertex. That removes through-bond cases from a through-space contact model.

For an accepted ring and vertex j ,

$$\mathbf{d}_{ij} = \mathbf{r}_i - \mathbf{v}_j, \quad r_{ij} = |\mathbf{d}_{ij}|, \quad \hat{\mathbf{d}}_{ij} = \frac{\mathbf{d}_{ij}}{r_{ij}}. \quad (6)$$

The distance r_{ij} is in Å, while $\hat{\mathbf{d}}_{ij}$ is unitless. A vertex is skipped when $r_{ij} < 0.1$ Å, the singularity guard in the parameter file, or when $r_{ij} > 5.0$ Å, the dispersion vertex cutoff.

Between 4.3 Å and 5.0 Å the code multiplies the contribution by a smooth switching function:

$$S(r) = \begin{cases} 1, & r \leq R_s, \\ \frac{(R_c^2 - r^2)^2 (R_c^2 + 2r^2 - 3R_s^2)}{(R_c^2 - R_s^2)^3}, & R_s < r < R_c, \\ 0, & r \geq R_c. \end{cases} \quad (7)$$

Here $R_s = 4.3 \text{ Å}$ is the switching onset and $R_c = 5.0 \text{ Å}$ is the outer cutoff. Both values are read from `data/calculator_params.toml`. The function and its first derivative match at both endpoints: the contribution is full strength before R_s , fades smoothly across the taper, and reaches zero at R_c . If the computed switch value is below 10^{-15} , the vertex is treated as zero.

The per-vertex tensor is

$$K_{ab}^{(ij)} = S(r_{ij}) \left(\frac{3 d_{ij,a} d_{ij,b}}{r_{ij}^8} - \frac{\delta_{ab}}{r_{ij}^6} \right) = \frac{S(r_{ij})}{r_{ij}^6} (3 \hat{d}_{ij,a} \hat{d}_{ij,b} - \delta_{ab}). \quad (8)$$

The indices a, b run over x, y, z . The Kronecker delta δ_{ab} is one when $a = b$ and zero otherwise. Since $d\mathbf{d}/r^8$ has units Å^{-6} , every entry of $\mathbf{K}^{(ij)}$ has units Å^{-6} . Its trace is zero: the three diagonal entries sum to

$$\frac{S(r)}{r^6} (3|\hat{\mathbf{d}}|^2 - 3) = 0.$$

The scalar contact stored beside it is

$$q^{(ij)} = \frac{S(r_{ij})}{r_{ij}^6}, \quad (9)$$

also in Å^{-6} .

A useful exact picture is the eigendecomposition of Eq. (8). Each accepted vertex contributes a signed uniaxial ellipsoid. Along the vertex-to-target direction $\hat{\mathbf{d}}$, the tensor has eigenvalue $2S/r^6$. In either perpendicular direction it has eigenvalue $-S/r^6$. The signed sum of those three eigenvalues is zero, so the tensor records anisotropy rather than volume. Moving the vertex farther away shrinks all three axes as r^{-6} , and the switching function scales all three axes together. Summing a ring means adding these oriented, traceless ellipsoids from its vertices.

For a ring R , the code accumulates

$$\mathbf{K}^{(iR)} = \sum_{j \in R_i} \mathbf{K}^{(ij)}, \quad q^{(iR)} = \sum_{j \in R_i} q^{(ij)}, \quad (10)$$

where R_i is the set of vertices of ring R that pass the distance, cutoff, taper, and numerical-floor tests for target i . If no vertex passes, the ring contributes nothing. Otherwise the ring tensor is decomposed with Eqs. (2)–(5); its T_2 components and scalar contact strength are added to the per-ring-type accumulators.

The atom total is then

$$\mathbf{K}_i = \sum_{R \in \mathcal{A}_i} \mathbf{K}^{(iR)}, \quad (11)$$

where $\mathcal{A}_i$ is the set of accepted rings with at least one accepted vertex for atom i . The code stores `SphericalTensor::Decompose( $\mathbf{K}_i$ )` in the field named `disp_shielding_contribution`. The name is historical shorthand in the data structure. The stored tensor is the uncalibrated Å^{-6} geometry from Eq. (11).

The same per-vertex expression is used when the result is sampled at an arbitrary point in space. That path skips points inside a ring-radius sphere, points whose ring center is farther than 15.0 Å, and vertices that fail the same 0.1 Å singularity guard, switching-floor test, or 5.0 Å vertex cutoff. It returns the decomposed tensor field at the sample point and does not produce the scalar contact output.

The main approximations are visible in these equations. The source sites are aromatic ring vertices, not all atoms in the protein. Each vertex has unit dispersion coefficient; element-specific C_6 values and polarizabilities are not multiplied into the tensor. The physical long-range r^{-6} tail is truncated, with a taper over the last 0.7 Å. The near-field and through-bond exclusions define where the through-space vertex model is not evaluated.

5 Inputs and output

This calculator consumes molecular geometry and topology: atom coordinates in Å, aromatic ring membership, ring vertex coordinates, ring centers and radii, and the covalent bond graph used for the through-bond exclusion. Its declared runtime dependencies are `GeometryResult` and `SpatialIndexResult`. It does not read MOPAC charges, AIMNet2 charges or embeddings, ff14SB charges, APBS electrostatic fields, or DSSP hydrogen-bond assignments; those are inputs to other calculators.

The calculator's own `WriteFeatures` method writes three arrays.

`disp_shielding.npy` has shape $(N, 9)$, one row per atom, in the `SphericalTensor` order given above; for this tensor the T_2 entries carry the angular information.

`disp_per_type_T0.npy` has shape $(N, 8)$, with one scalar $q = \sum S/r^6$ contact sum per aromatic ring type. Despite the file suffix, this scalar is not the trace of $\mathbf{K}$.

`disp_per_type_T2.npy` has shape $(N, 40)$, with five T_2 components for each of the eight aromatic ring-type blocks. All three outputs are geometric quantities in Å⁻⁶. Calibration against the DFT reference is what turns these descriptors into a shielding contribution; before that step they should be read as geometry that may correlate with the dispersion part of shielding.